CAVENDISH UNIVERSITY ZAMBIA

BsC COMP

Research/Project Proposal

FACULTY: Business, Information Technology and Professional Studies

RESEARCH/PROJECT TITLE: Design and Development of MindSanctuary: A University-Centric Wellbeing and Anonymous Peer Support Platform

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CHAPTER ONE: INTRODUCTION

1.0 Introduction

The transition to university life introduces numerous psychological and social challenges for students, including academic pressure, financial strain, and social isolation. Although universities provide counselling services, these are often overstretched and hindered by stigma, discouraging help-seeking behaviors. This proposal presents *MindSanctuary*—a web-based platform designed to offer accessible, immediate, and non-clinical wellbeing support for university students. By integrating mood tracking, a wellbeing resource hub, and anonymous peer-to-peer chat, *MindSanctuary* aims to foster a safe digital environment where students can proactively manage their mental health.

1.1 Background

Globally, the prevalence of mental health issues among university students is a growing concern. According to the World Health Organization (WHO, 2022), approximately one in five young adults experience anxiety or depression symptoms. Universities in developed nations have begun adopting digital mental health interventions, such as therapy apps and online counselling portals, to supplement traditional services (Lui et al., 2021). These platforms leverage technology to provide scalable, 24/7 support, reducing barriers to access.

In the local Zambian context, and at Cavendish University specifically, mental health support structures, while present, face challenges of resource limitations and persistent cultural stigma associated with seeking psychological help (Mweemba & Mwale, 2023). Many students may be reluctant to approach formal services due to fears of being judged. This creates a significant gap between the need for support and its accessibility. There is a clear lack of platforms that offer immediate, low-threshold support which prioritizes user anonymity and peer connection, bridging the gap before a student seeks professional help. MindSanctuary is designed to address this specific gap by providing a locally-relevant, university-centric platform that empowers students and volunteers within the community to support one another.

1.2 Motivation of the Study

The motivation for this study stems from a recognition of the silent struggle many students endure. Witnessing peers grapple with stress and isolation without accessible outlets for support highlights a critical need. The researcher is driven by the potential of technology to create compassionate, scalable solutions. Developing MindSanctuary is an opportunity to apply software engineering skills to a pressing social issue, creating a tangible tool that can positively impact the student community at Cavendish University and potentially beyond.

1.3 Significance of the Study

This project is significant for several reasons. For students, it provides an accessible, anonymous, and immediate channel for emotional support and wellbeing resource access. For the university administration, it offers a scalable supplement to existing counselling services and provides valuable, aggregated insights into wellbeing trends of the student population. For the academic community, it serves as a practical implementation of a modern, real-time web application using robust technologies like Elixir and Phoenix, demonstrating the application of software engineering principles to solve real-world problems.

1.4 Scope of the Study

The scope of this study is confined to the design, development, and testing of a functional web application prototype. The platform will include user authentication, a mood tracker, a resource hub for articles and audio, and a one-to-one anonymous chat system connecting students with trained volunteer peers. The study will not include clinical diagnosis, long-term counselling, integration with official university SSO systems, or the development of native mobile applications, though the design will be responsive for mobile browsers.

1.5 Problem Statement

University students at Cavendish University often face significant psychological distress but are hindered from seeking support due to the stigma associated with mental health services

and the limited capacity of existing counselling resources. The absence of an immediate, anonymous, and accessible platform for non-clinical peer support and wellbeing management exacerbates this situation, potentially leading to worsened academic performance and personal wellbeing.

1.6 Objectives

1.6.1 General Objective

To design and develop a secure, web-based wellbeing platform named "MindSanctuary" that provides university students with tools for mood tracking, access to wellbeing resources, and anonymous real-time peer support.

1.6.2 Specific Objectives

1. To analyze and specify the functional and non-functional requirements for the MindSanctuary platform, focusing on security, anonymity, and usability.
2. To design the system architecture, data models, and user interface prototypes for the core modules: user management, mood tracking, resource hub, and anonymous chat.
3. To implement a fully functional web application using the Elixir programming language and the Phoenix framework, incorporating user authentication, a mood tracker, a resource hub, and a real-time chat system.
4. To rigorously test the implemented system for functionality, security vulnerabilities, and user acceptance to ensure it meets the specified requirements.

1.7 Research Questions

1. What are the core functional and non-functional requirements necessary to build a secure and effective university-based wellbeing and anonymous peer support platform?
2. How can a system architecture be designed to effectively support real-time anonymous communication, data persistence for wellbeing tracking, and robust user management while ensuring data privacy?

3. What are the key technical challenges in implementing a real-time, anonymous chat system with Phoenix Channels and ensuring its integration with other platform modules like mood tracking and resource management?
4. To what extent does the developed MindSanctuary platform meet the criteria of functionality, security, and usability as perceived through structured testing protocols?

1.8 Conceptual/Theoretical Framework

This project is grounded in the **Technology Acceptance Model (TAM)** (Davis, 1989), which posits that the perceived usefulness and perceived ease of use of a system determine its adoption. MindSanctuary is designed with these factors in mind: its usefulness is derived from providing immediate support and resources, while its ease of use is ensured through a simple interface and anonymous access, lowering the barrier to use. Furthermore, the design incorporates principles from **Online Community Moderation** theories (Zhu et al., 2021) to create a safe environment, using profanity filtering and reporting mechanisms to manage user interactions and mitigate potential harm.

1.9 Definition of Terms

Phoenix LiveView: A library for the Elixir programming language that enables rich, real-time user experiences with server-rendered HTML.

Phoenix Channels: A technology within the Phoenix framework that allows for soft-realtime communication via WebSockets, used for building features like chat.

Oban: A robust job processing library for Elixir, used for running background tasks like matching chat volunteers.

Ephemeral Pseudonym: A temporary, randomly generated username used in chat to protect the real identity of the user.

Non-clinical Support: Support that is geared towards general wellbeing and is not a substitute for professional psychotherapy or psychiatric treatment.

1.10 Organization of the Report

The report is organized into chapters. Chapter One introduces the study. Chapter Two reviews relevant literature on digital mental health interventions and real-time web technologies. Chapter Three details the research methodology and system design, including architecture and data models. Subsequent chapters will cover the system implementation, testing results, and a conclusion with recommendations for future work.

CHAPTER TWO: LITERATURE REVIEW

2.0 Overview

This chapter explores existing research and technological work relevant to the MindSanctuary project. It examines the landscape of digital mental health tools, the efficacy of peer support models, and the technical architectures used to build modern, real-time web applications. The goal is to situate this project within the current body of knowledge and identify a clear gap that MindSanctuary aims to fill.

2.1 Literature Review

The global push towards digital mental health solutions has been accelerated by the increased recognition of mental health needs and advances in technology. Studies have shown that digital interventions, particularly those based on Cognitive Behavioral Therapy (CBT) principles, can be effective in reducing symptoms of anxiety and depression (Lui et al., 2021). Furthermore, the role of peer support in recovery and wellbeing is well-established. Peer support offers experiential empathy and understanding that can reduce feelings of isolation (Shalaby & Agyapong, 2020).

The integration of these two domains—digital technology and peer support—has given rise to online peer support platforms. These platforms vary from forum-based communities to real-time chat services. A key challenge identified in the literature is maintaining user safety and privacy, especially in anonymous settings (Pfeiffer et al., 2020). Effective moderation, clear community guidelines, and technical features like reporting and blocking are critical success factors. This underscores the need for MindSanctuary to incorporate robust moderation tools from the outset.

2.2 Related Works

Several existing platforms share conceptual similarities with MindSanctuary. **7 Cups of Tea** is a prominent example, offering free, anonymous emotional support through listener-volunteers via chat. However, it is a general-purpose platform not tailored to the specific context and resources of a university like Cavendish. **Talkspace** and **BetterHelp** provide professional therapy, a clinical service which is outside the scope of this project and often involves a cost barrier.

From a technical perspective, modern web applications like **Discord** and **Slack** demonstrate the power of Phoenix-like technologies (or their equivalents in other languages) for building real-time, scalable communication systems. These platforms show that Elixir's Phoenix Framework, with its built-in support for Channels and Presence, is a technologically sound choice for the chat component of MindSanctuary (Tate, 2025). The use of Elixir also promotes high reliability and maintainability, which are crucial for a support application.

2.3 Gaps in the Literature

While literature on digital mental health is abundant, there is a scarcity of research focusing on the development and implementation of **university-specific, non-clinical, anonymous peer-to-peer chat platforms** in the Zambian context. Most studies evaluate existing, generalized platforms or clinical telehealth services. Furthermore, technical papers detailing the architecture of such systems with a strong emphasis on data anonymity and privacy-by-design are limited. This project will contribute by detailing the end-to-end development of a context-aware platform and documenting the architectural decisions and security measures taken to create a safe and effective support tool for a specific student population.

CHAPTER THREE: RESEARCH METHODOLOGY AND SYSTEM DESIGN

3.0 Overview

This chapter describes the methodology that will be employed to achieve the project's objectives. A combination of research and development (R&D) and agile software development methodologies will be used. The system design is presented through architectural diagrams and data models to provide a blueprint for implementation.

3.1 Research Design

The study will adopt a **Design and Creation** research strategy. This approach is ideal for information systems projects where the primary outcome is the creation of a novel IT artifact—in this case, the MindSanctuary platform. The process will be iterative, involving cycles of design, development, and testing, aligning with agile principles to accommodate feedback and changes.

3.2 Baseline Study

3.2.1 Data Collection

Data collection during the requirements analysis phase (Objective 1) will be primarily through the review of existing literature (as in Chapter 2) and the analysis of similar platforms. A simple questionnaire may be distributed to a small group of students to gather initial feedback on desired features and usability preferences. The primary "tool" for development will be the Elixir/Phoenix codebase itself.

3.2.2 Research Approach

A **pragmatic** approach will be used, focusing on practical solutions to the identified problem. The research is applied in nature, aiming to produce a working software system that solves a real-world problem.

3.2.3 Development of the Application

The application will be developed using the Elixir programming language and the Phoenix web framework. The development will follow an iterative process:

1. **Sprint 1:** Project setup, user authentication, and basic mood tracker.
2. **Sprint 2:** Resource hub CRUD operations and listing.
3. **Sprint 3:** Core chat functionality (Channels, Message Persistence).
4. **Sprint 4:** Volunteer dashboard, pairing logic, and moderation tools.
5. **Sprint 5:** Integration, testing, and deployment.

Version control will be managed using Git, hosted on GitHub for collaborative development.

3.2.4 System Design

The system will follow a Model-View-Controller (MVC) pattern, as implemented by Phoenix, with LiveView handling the View-Controller layer for a highly interactive experience.

3.3 System Design

3.3.1 Context Diagram

The context diagram below shows the system's boundaries and its interactions with external entities.

```
+-------------------+     (Manages Resources, Views Reports)     +-------------------+
|                   | <----------------------------------------> |                   |
|   Administrator   |                                           | MindSanctuary     |
|                   |                                            | System            |
+-------------------+                                            +-------------------+
                                                                  ^              ^
                                                                  |              |
                 (Requests Chat, Sends Messages)                  |              |
                                                                  |              |
                                                                  |              |
+-------------------+    (Tracks Mood, Views Resources)          |              |    (Accept
s Chats, Provides Support)
|                   | <----------------------------------------> |              |
<----------------------------------------> +-------------------+
|      Student      |                                            |              |
|    Volunteer      |
|                   |                                            |              |
|                   |
+-------------------+                                            +-------------------+
+-------------------+
```

3.3.2 System Software Level Architectural Design

The system will use a Monolithic architecture with modular contexts Phoenix application structure. The key components are:

- **Web Layer:** Phoenix Router, Controllers, and LiveViews.
- **Business Logic Layer:** Phoenix Contexts (Accounts, Wellbeing, Support).
- **Data Layer:** Ecto for PostgreSQL database interaction.
- **Real-Time Layer:** Phoenix Channels and Presence over WebSockets.
- **Background Processing:** Oban for job queues (e.g., chat pairing).
- **Mini-IO Services:** Ex Aws S3 for document/media file storage.

3.3.3 Modular Design of the System Functions

The system is decomposed into the following modules, aligned with Phoenix Contexts:

- **Accounts Context:** Handles user registration, authentication, and profile management.
- **Wellbeing Context:** Manages mood entries and resource hub data (articles, audio).
- **Support Context:** Orchestrates the entire chat flow—queue management, message persistence, volunteer pairing, and reporting.

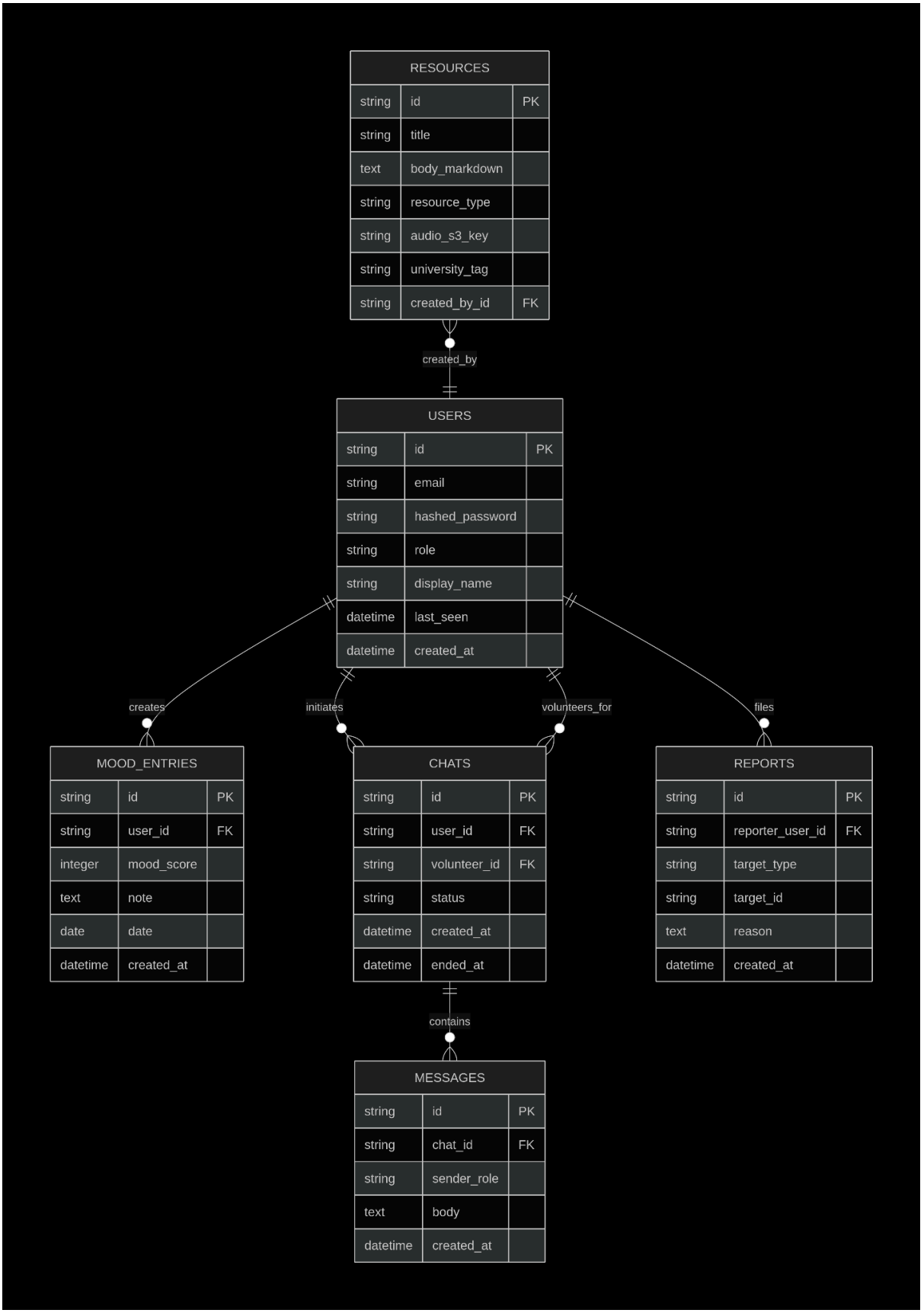
3.3.4 System Class Diagram

(A simplified version focusing on core entities)

```
+-------------+       +-------------+       +-------------+
|    User     |       |  MoodEntry  |       |  Resource   |
+-------------+       +-------------+       +-------------+
| - id        |       | - id        |       | - id        |
| - email     |       | - user_id   |       | - title     |
| - role      |       | - score     |       | - type      |
| - name      |       | - note      |       | - s3_key    |
+-------------+       +-------------+       +-------------+
      ^                     ^                     ^
      |                     |                     |
      | 1                  *| 1                  *| 1
      | *                   |                     |
+-------------+       +-------------+       +-------------+
|    Chat     |       |   Message   |       |   Report    |
+-------------+       +-------------+       +-------------+
| - id        |       | - id        |       | - id        |
| - student_id|       | - chat_id   |       | - target_id |
| - volunteer_|       | - body      |       | - reason    |
| - status    |       | - sender_role|      | - resolved_at|
+-------------+       +-------------+       +-------------+
```

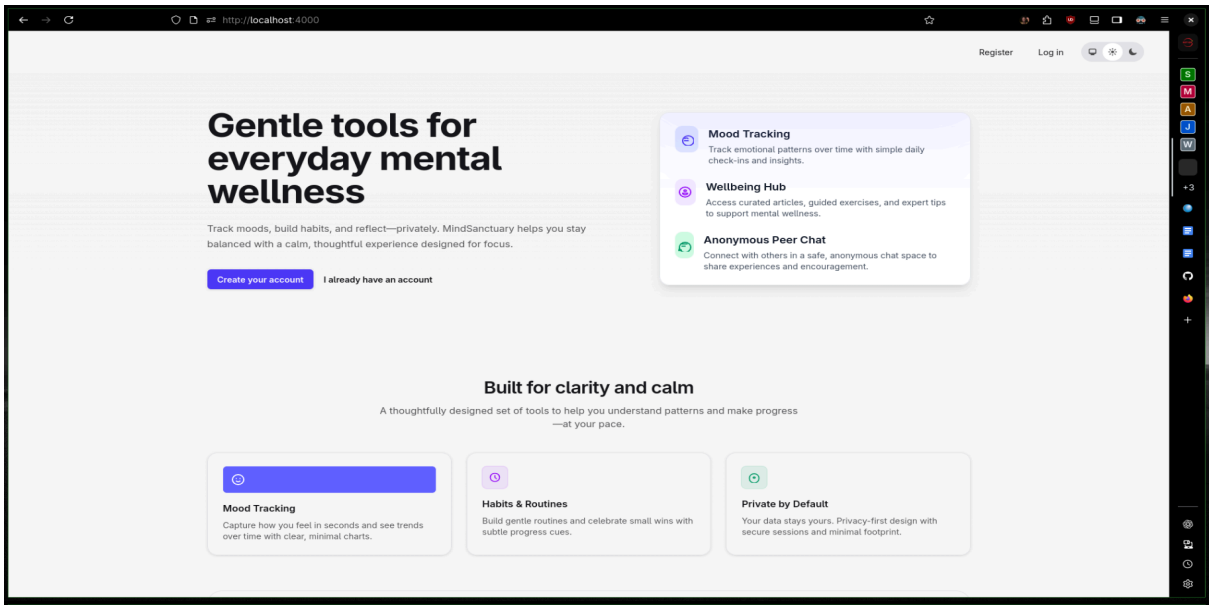
3.3.5 System Data Model Design

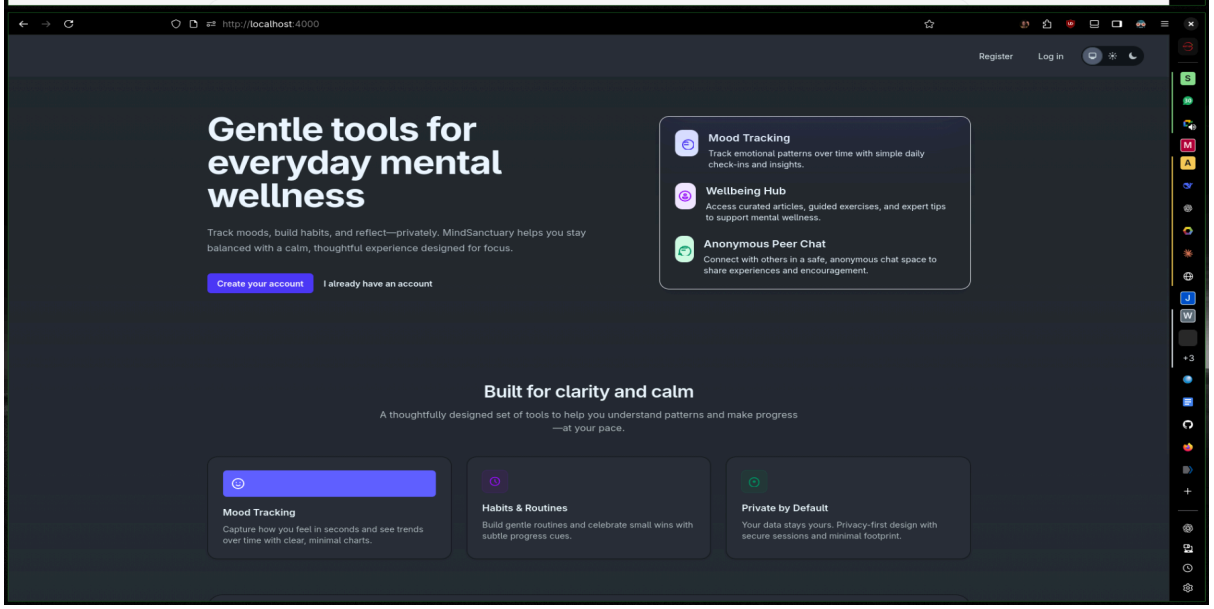
The Entity-Relationship Diagram (ERD) below visualizes the core data model described in the initial technical specification, showing the relationships between users, mood entries, chats, messages, and resources.

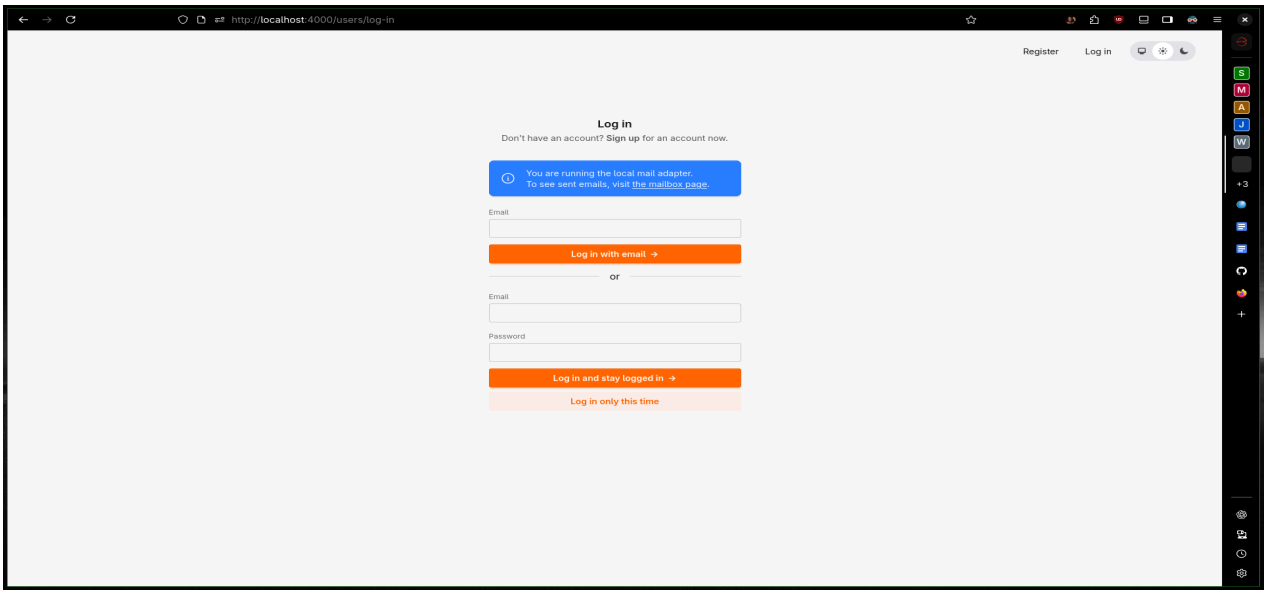

RESOURCES
string id PK
string title
text body_markdown
string resource_type
string audio_s3_key
string university_tag
string created_by_id FK
created_by
USERS
string id PK
string email
string hashed_password
string role
string display_name
datetime last_seen
datetime created_at
creates
initiates
volunteers_for
files
MOOD_ENTRIES
string id PK
string user_id FK
integer mood_score
text note
date date
datetime created_at
CHATS
string id PK
string user_id FK
string volunteer_id FK
string status
datetime created_at
datetime ended_at
REPORTS
string id PK
string reporter_user_id FK
string target_type
string target_id
text reason
datetime created_at
contains
MESSAGES
string id PK
string chat_id FK
string sender_role
text body
datetime created_at

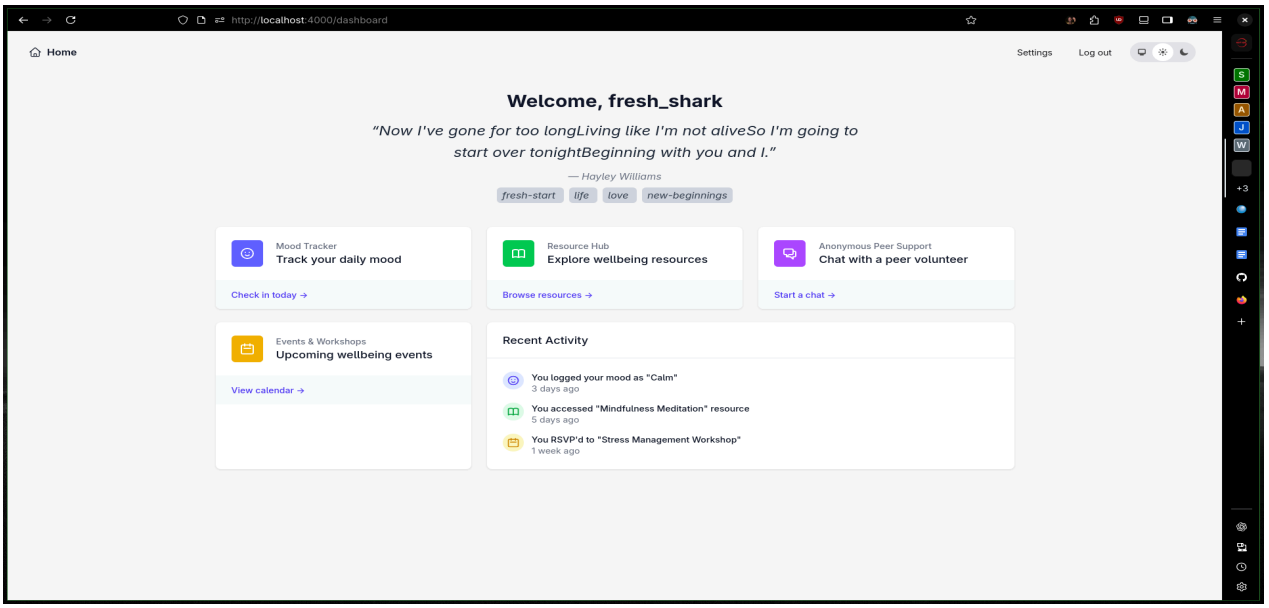
CHAPTER FOUR: USER INTERFACE AND USER EXPERIENCE (UI/UX)

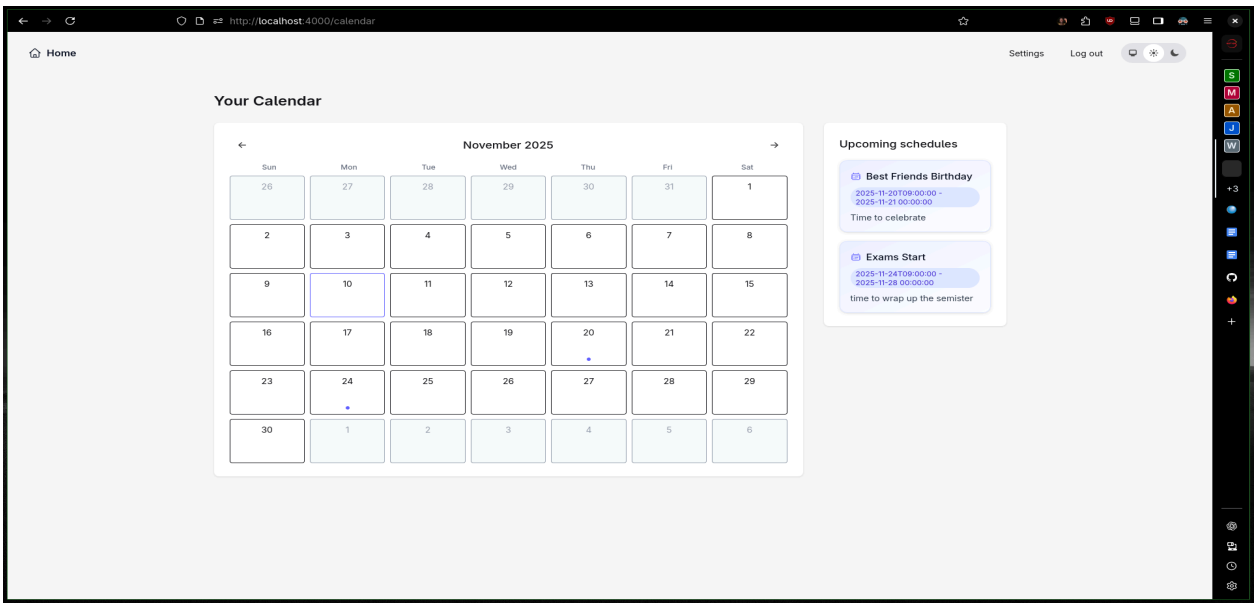
This chapter presents the visual design and interaction flow of the MindSanctuary platform. The user interface (UI) focuses on simplicity, accessibility, and emotional comfort, ensuring that students can navigate and engage with the system intuitively whilst allowing user choice to a dark or light interface. Each screen was designed to align with the project's wellbeing objectives—promoting calmness, privacy, and ease of use. The following screenshots illustrate key interfaces of the platform, including user authentication, mood tracking, the resource hub, and the anonymous peer support chat.

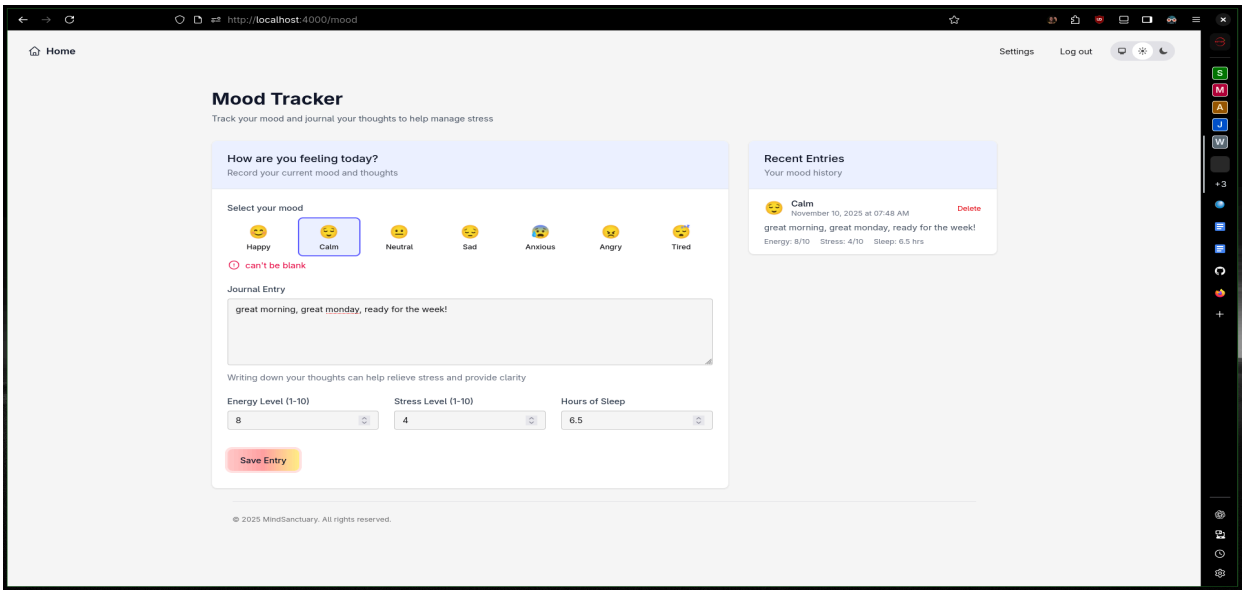


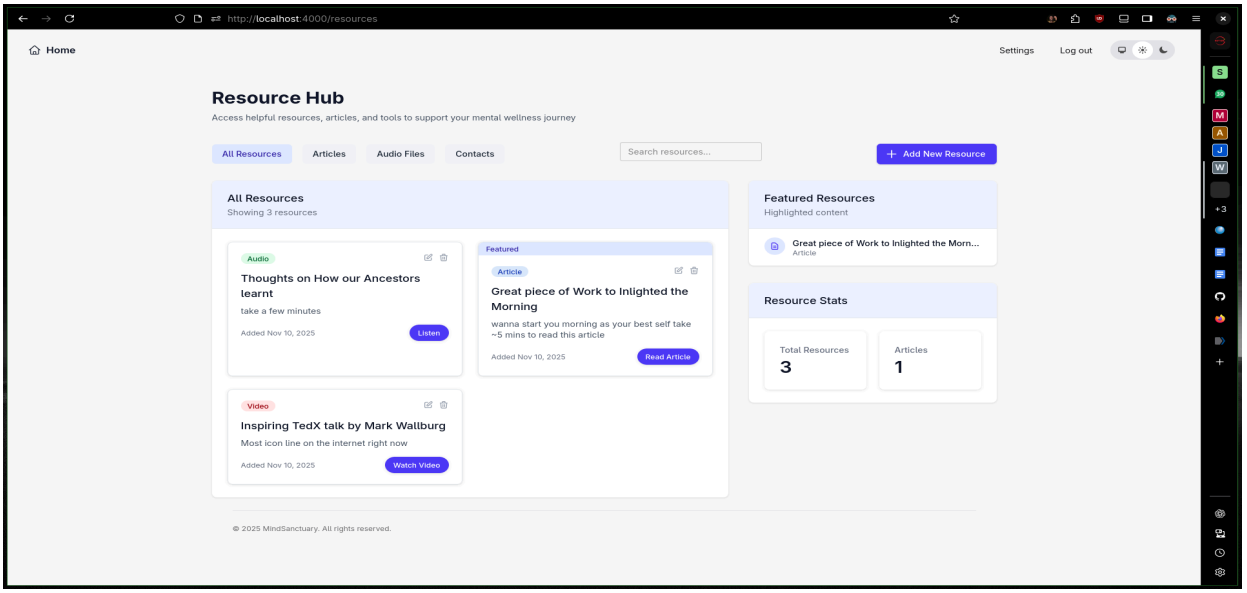



http://localhost:4000/users/log-in
Register
Log in
Log in
Don't have an account? Sign up for an account now.
You are running the local mail adapter.
To see sent emails, visit the mailbox page.
Email
Log in with email →
or
Email
Password
Log in and stay logged in →
Log in only this time


http://localhost:4000/dashboard
Home
Settings
Log out
Welcome, fresh_shark
"Now I've gone for too longLiving like I'm not aliveSo I'm going to start over tonightBeginning with you and I."
— Hayley Williams
fresh-start
life
love
new-beginnings
Mood Tracker
Track your daily mood
Check in today →
Resource Hub
Explore wellbeing resources
Browse resources →
Anonymous Peer Support
Chat with a peer volunteer
Start a chat →
Events & Workshops
Upcoming wellbeing events
View calendar →
Recent Activity
You logged your mood as "Calm"
3 days ago
You accessed "Mindfulness Meditation" resource
5 days ago
You RSVP'd to "Stress Management Workshop"
1 week ago


http://localhost:4000/calendar
Home
Settings
Log out
Your Calendar
November 2025
Sun
Mon
Tue
Wed
Thu
Fri
Sat
Upcoming schedules
Best Friends Birthday
2025-11-20T09:00:00 - 2025-11-21 00:00:00
Time to celebrate
Exams Start
2025-11-24T09:00:00 - 2025-11-28 00:00:00
time to wrap up the semister


http://localhost:4000/mood
Home
Settings
Log out
Mood Tracker
Track your mood and journal your thoughts to help manage stress
How are you feeling today?
Record your current mood and thoughts
Select your mood
Happy
Calm
Neutral
Sad
Anxious
Angry
Tired
can't be blank
Journal Entry
great morning, great monday, ready for the week!
Writing down your thoughts can help relieve stress and provide clarity
Energy Level (1-10)
8
Stress Level (1-10)
4
Hours of Sleep
6.5
Save Entry
Recent Entries
Your mood history
Calm
November 10, 2025 at 07:48 AM
Delete
great morning, great monday, ready for the week!
Energy: 8/10
Stress: 4/10
Sleep: 6.5 hrs
© 2025 MindSanctuary. All rights reserved.


http://localhost:4000/resources
Home
Settings
Log out
Resource Hub
Access helpful resources, articles, and tools to support your mental wellness journey
All Resources
Articles
Audio Files
Contacts
Search resources...
Add New Resource
All Resources
Showing 3 resources
Audio
Thoughts on How our Ancestors learnt
take a few minutes
Added Nov 10, 2025
Listen
Featured
Article
Great piece of Work to Inlighted the Morning
wanna start you morning as your best self take ~5 mins to read this article
Added Nov 10, 2025
Read Article
Video
Inspiring TedX talk by Mark Wallburg
Most icon line on the internet right now
Added Nov 10, 2025
Watch Video
Featured Resources
Highlighted content
Great piece of Work to Inlighted the Morn...
Article
Resource Stats
Total Resources
3
Articles
1
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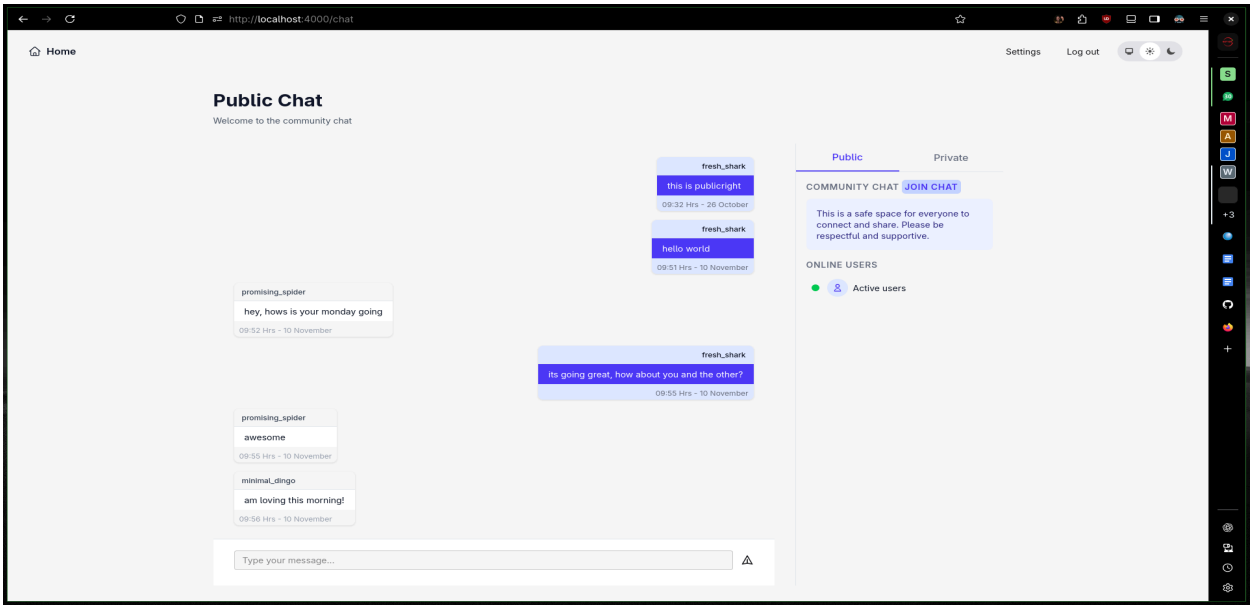
http://localhost:4000/chat
Home
Settings
Log out
Public Chat
Welcome to the community chat
fresh_shark
this is publicright
09:32 Hrs - 26 October
fresh_shark
hello world
09:51 Hrs - 10 November
promising_spider
hey, hows is your monday going
09:52 Hrs - 10 November
fresh_shark
its going great, how about you and the other?
09:55 Hrs - 10 November
promising_spider
awesome
09:55 Hrs - 10 November
minimal_dingo
am loving this morning!
09:56 Hrs - 10 November
Type your message...
Public
Private
COMMUNITY CHAT
JOIN CHAT
This is a safe space for everyone to connect and share. Please be respectful and supportive.
ONLINE USERS
Active users

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